# AWS-VPC-Peering

Project Defense Notes

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1.Why is the return route needed?

Ans: In this demonstrative project, When APP-B sends a packet to APP-A throught VPC peering the packet can reach APP-A if only the forward route is correctly configured. The APP-A also needs a route  to send the response back to APP-B that is the reason we need a return route.without it the request might reach its destination but response might not reach to the original sender.

VPC peering does not auto add routes to your tables we must config on both sides both need valid routing/ VPC-A(10.20.0.0/16)----------VPC-B(10.10.0.0/16)

2.Can VPC-B use VPC-As internet gateway over peering?

Ans: Apparently NO VPC-B cannot use VPC-A peering. VPC peering is designed for private communication between resources in the connected VPC it is not for sharing internet gateway nor providing transitional routing. In this project VPC-B has a private subnet and no internet gateway of its own If VPC-B need internet access we would need to create a separate solution like NAT gateway in an appropriately config network

VPC-A(internet gateway for public subnet)

VPC-B( communicate privately with VPC-A via peering)

3.Why is the APP-B SSH source the bastion private IP?

Ans:  In this project we used Bastion host as the controlled entry point for administrative access. Instead of allowing SSh connections from anywhere APP-B security group can restrict SSH access to the bastion hosts private IP address this reduces the attack surface because only traffic originating from that specific private IP is permitted by that rule. The admin connects to bastion first the —- use it to access private resources

The bastion private IP 10.10.1.10 so security group can only allow TCP port 22 from 10.10.1.10/32 so its a least privilege access

4. Why does active peering not guarantee a successful ping?

Ans: Active peering status only tells us that the VPC peering Connection has been established. It doesnt automatically config all the networking and security settings required for communication. For a successful ping we need the correct route in both VPC security group rules allowing ICMP traffic and suitable network ACL rules in any packets get blocked or replies the ping can fail even when the peering connection is still active

If ping fails we should check: peering connection status/forward return route in both route table/destination inbound ICMP security group Rules/Network ACL rules allowing request and reply

## ***Summary***:-

In this AWS VPC Peering project we configured return routes to ensure bidirectional communication between VPCs, restricted SSH access to the bastion host's private IP for better security, and verified that successful connectivity requires correct routing and security rule not just an Active peering status. VPC peering enables private communication between VPCs but does not allow VPC B to use VPC A's internet gateway or provide transitive internet access.////